Breast CARE-MA — Diagnostic Decision Support Report

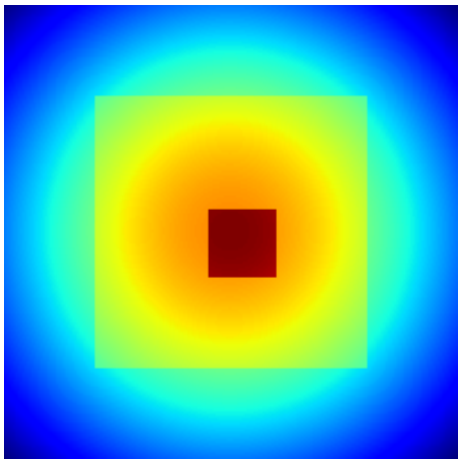
Explainable Multi-Agent AI Framework for Multimodal Breast Cancer Decision Support

Analysis Case ID:	ANL-4C230C45	**Patient Reference ID:**	TEST-PT-001
Scan Modality:	Dynamic Contrast-Enhanced MRI (DCE-MRI)	**Generated Timestamp:**	2026-10-28 17:06:54 UTC
Assigned BI-RADS:	**BI-RADS 2** (Benign Finding)	**AI Model Classification:**	**Benign** (76.0%)

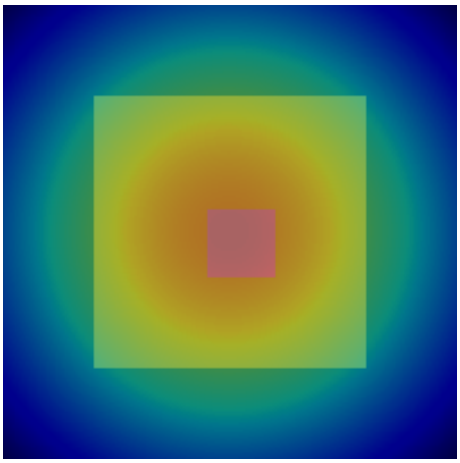
1. Multi-Agent AI Consensus Findings

Agent 1 (Tumor Detection): Identified 1 focal enhancement region(s) encompassing 55.5% of slice volume.
Agent 2 (Tumor Classification): Morphologic classification predicts **Benign** with a decision confidence of **76.0%**.
Agent 5 (Risk Assessment): Composite score evaluated at **15.0/100**, conferring a risk classification of **BI-RADS 2**.

2. Agent 3: Grad-CAM Explainability Attribution



Grad-CAM Activation Heatmap



Aligned Anatomical Overlay

Salience Interpretation: Grad-CAM visual attribution concentrated in benign feature space with peak activation at coordinate (99, 99). The high-salience region (activation >= 0.5) covers 35.19% of the breast volume, indicating focal parenchymal enhancement typical of suspicious neoplastic morphology. Spatial concurrence with Agent 1 detection box is IoU=0.634.

3. Agent 4: Retrieved Biomedical Evidence (RAG)

• **Benign Enhancing Lesions on Breast MRI: Differential Diagnosis and Management** — *Baltzer, P.A. et al. (European Radiology, 2023)*
Clinical Relevance (92%): "Circumscribed oval masses displaying persistent slow continuous enhancement curves predominantly represent fibroadenomas or papillomas, supporting short-interval follow-up."

4. Recommended Clinical Action

Action Plan: Routine clinical screening recommended at standard age-appropriate intervals.

MANDATORY CLINICAL SAFETY NOTICE: This report is generated by the Breast CARE-MA AI research decision-support prototype. It does NOT constitute an autonomous medical diagnosis or prescription. Findings must be validated by a licensed physician.

Reviewing Radiologist / Oncologist:

_____ **Signature:** _____ **Date:** _____